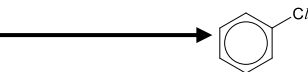
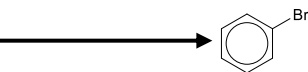
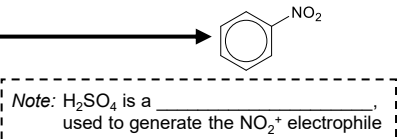


Reactions

(electrophilic substitution)



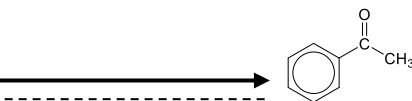
Generation of electrophile:

Note: AlCl_3 is a _____, used to generate the Cl^+ electrophile

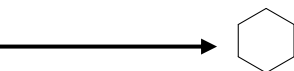
electrophile:



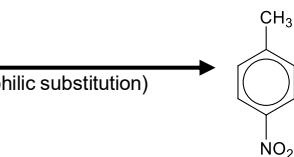
electrophile:



(reduction)



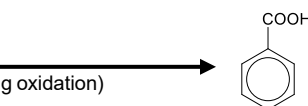
(electrophilic substitution)



other electrophilic substitution

free-radical substitution

(strong oxidation)



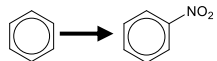
Why electrophilic substitution?

- Explain why arenes undergo electrophilic substitution rather than electrophilic addition
- The _____ in benzene ring gives rise to resonance stabilisation
- Arenes do not undergo electrophilic addition because it disrupts the _____
- For electrophilic substitution, while the resonance stabilisation is _____, it is _____ in the product

Mechanism

Name: _____

Generation of electrophile:

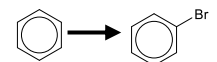


Step 1:

Step 2:

Name: _____

Generation of electrophile:



Step 1:

Step 2:

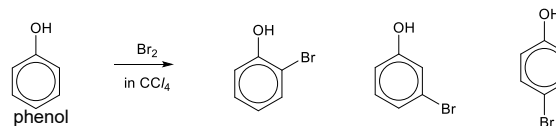
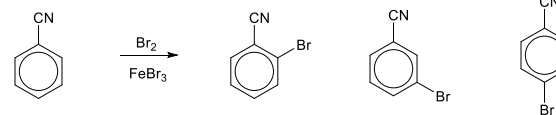
Effects of substituents

1. Reactivity

Example	explanation on reactivity
	The $-\text{CH}_3$ group is <u>electron-donating</u> which <u>increases the electron density</u> of the benzene ring, making it <u>more susceptible</u> towards electrophilic attack; i.e. attack by electrophiles.
	The $-\text{OH}$ group is <u>electron-donating</u> which _____ the electron density of the benzene ring, making it _____ towards attack by electrophiles. <i>Note:</i> the $-\text{OH}$ group is electron-donating because orbital containing lone pair of electrons on O overlaps with the π electron cloud of the benzene ring, hence the _____ on O are able to _____ into the benzene ring
	The $-\text{NO}_2$ group is electron-_____ which _____ the electron density of the benzene ring, making it _____ towards attack by electrophiles.

2. Directing effect

Label the major and minor products

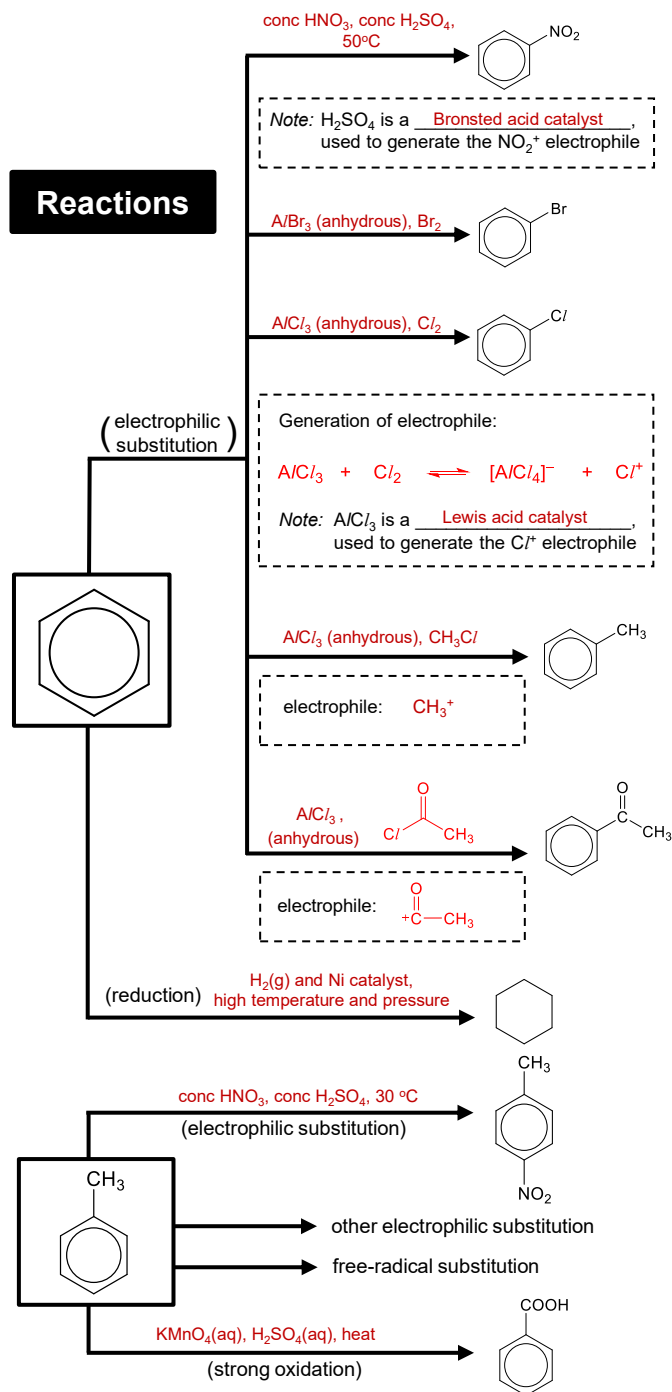


Note:
milder conditions can be used for the electrophilic substitution of phenol b/c the $-\text{OH}$ group makes it more reactive towards electrophilic substitution

- Suggest the reagent and conditions for the following 2 step synthesis.



Reactions



Why electrophilic substitution?

- Explain why arenes undergo electrophilic substitution rather than electrophilic addition
- The delocalisation of 6 π electrons in benzene ring gives rise to resonance stabilisation
- Arenes do not undergo electrophilic addition because it disrupts the resonance stabilisation
- For electrophilic substitution, while the resonance stabilisation is temporarily disrupted, it is restored in the product

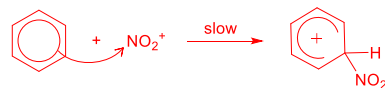
Mechanism

Name: electrophilic substitution

Generation of electrophile:



Step 1:



Step 2:

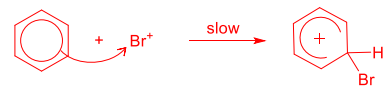


Name: electrophilic substitution

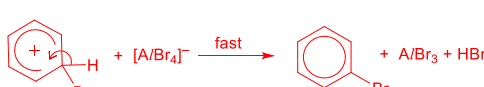
Generation of electrophile:



Step 1:



Step 2:



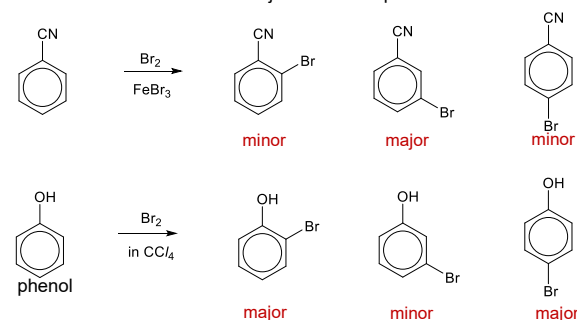
Effects of substituents

1. Reactivity

Example	explanation on reactivity
<chem>Cc1ccccc1</chem>	The $-\text{CH}_3$ group is <u>electron-donating</u> which <u>increases the electron density</u> of the benzene ring, making it <u>more susceptible</u> towards electrophilic attack; i.e. attack by electrophiles.
<chem>Oc1ccccc1</chem>	The $-\text{OH}$ group is <u>electron-donating</u> which <u>increases</u> the electron density of the benzene ring, making it <u>more susceptible</u> towards attack by electrophiles. <i>Note:</i> the $-\text{OH}$ group is electron-donating because orbital containing lone pair of electrons on O overlaps with the π electron cloud of the benzene ring, hence the <u>lone pair of electrons</u> on O are able to <u>delocalise</u> into the benzene ring
<chem>[O-][N+](=O)c1ccccc1</chem>	The $-\text{NO}_2$ group is electron- <u>withdrawing</u> which <u>decreases</u> the electron density of the benzene ring, making it <u>less susceptible</u> towards attack by electrophiles.

2. Directing effect

Label the major and minor products



Note: milder conditions can be used for the electrophilic substitution of phenol b/c the $-\text{OH}$ group makes it more reactive towards electrophilic substitution

➤ Suggest the reagent and conditions for the following 2 step synthesis.

